Supplementary Material

Temporal Retiming Attacks on Hybrid Quantum Spiking Classifiers

Authors withheld for review

This supplement preserves completed supporting experiments and the full paired timestamp comparison. Historical studies are not additional final-model N-MNIST robustness evaluations. Every section states the model, partition and evidence boundary needed for interpretation. Numerical values are transcribed from preserved reports; model training, attacks and original independent audits were not rerun during preparation. Source identifiers [R1]–[R10] are mapped to exact repository paths in Section S12.

S1 Development of the hybrid architecture

The selected development configurations are summarized in Table S1 and Figure S1. The earlier eight-dimensional frontend differs from the final 32-dimensional `wide4` frontend. Different training stages, continuation histories and budgets prevent causal attribution of the validation changes to a single circuit component. These are validation-only model-development results, rather than a controlled ablation establishing quantum advantage [R1].

Table S1. Selected quantum-head and frontend development results.

Configuration	Validation accuracy (%)	Parameters	Scope
Quantum v1	86.16	7,026	Earlier 8D frontend
Selected quantum head v2	92.60	7,130	Earlier 8D frontend
`wide4` classical control	97.80	38,122	32D frontend; seed 42
Final QSNN-v3	97.06	40,762	32D frontend; seed 42

Sources: [R1] and companion frontend/quantum-head development reports. Configuration changes and unequal training stages preclude a component-wise causal interpretation.

S1.1 Frozen final-model training details

All five QSNN-v3 runs share one hash-verified seed-42 pretrained frontend and one stratified 55,000/5,000 training/validation partition. The frontend is frozen for the first two epochs and then optimized while its BatchNorm layers remain in evaluation mode. Architecture and hyperparameters are frozen across seeds 42, 123, 777, 2026 and 6543. The official test partition is not instantiated for the final hybrid campaign [R1].

AdamW uses weight decay 10^{-4}, with learning rates 5×10^{-5} for the extractor, 2.5×10^{-4} for latent projection and circuit, and 10^{-3} for the classifier. Training uses batch size 256, a maximum of 40 epochs and gradient clipping at norm 1. The ReduceLROnPlateau schedule uses factor 0.5, patience 2 and minimum learning rate 10^{-5}. Checkpoint selection prioritizes validation accuracy and then validation loss; early stopping has patience 6 after at least eight epochs. The frontend uses float16 autocast, while circuit inputs are float32 and state amplitudes are complex64. These choices describe the simulated implementation and imply no quantum speedup.

The selected epochs are 37, 24, 40, 38 and 40; recorded runtimes are 1,200.00, 938.80, 1,245.30, 1,251.55 and 1,237.35 s, respectively. The seed-42 process was externally terminated during epoch 38, after saving its epoch-37 best checkpoint; that checkpoint was reloaded and evaluated for the reported result. Between-seed variability is conditional on shared pretraining and split, rather than variability across independently pretrained pipelines.

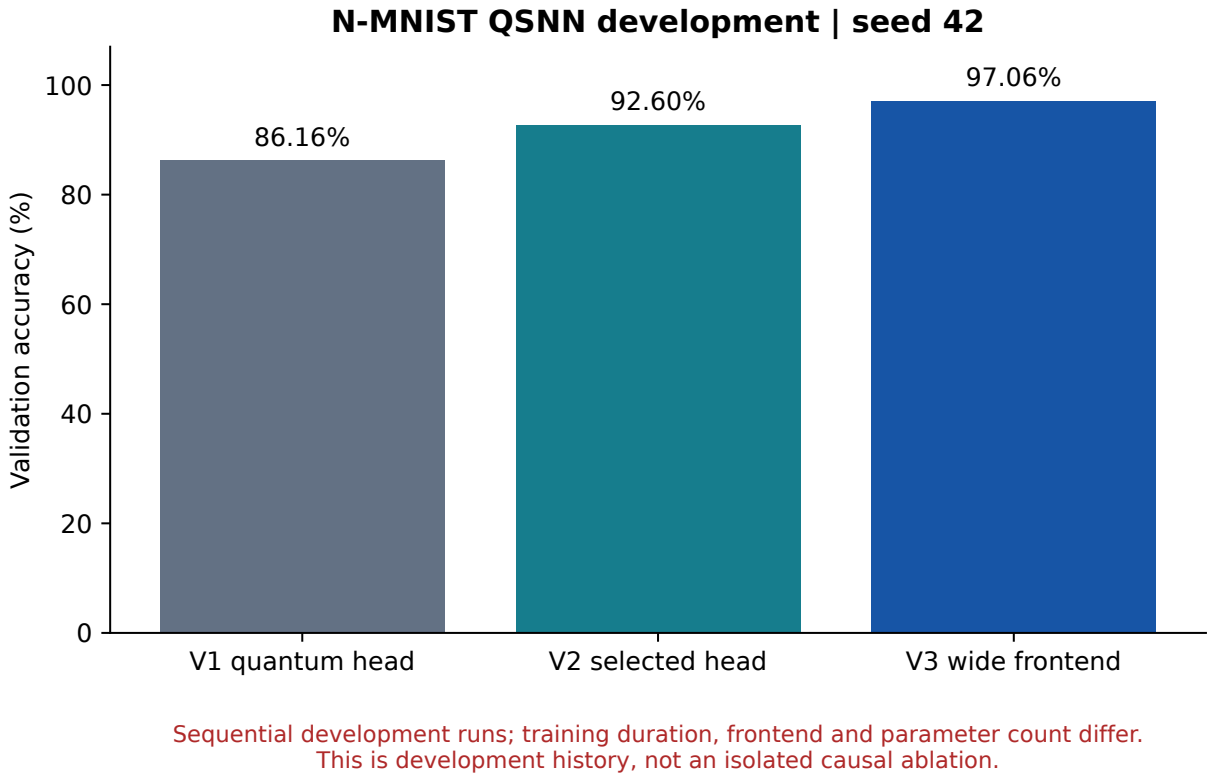


Figure S1. Validation-only development trajectory. Runs differ in frontend dimensions and training stages. The plot documents model development and is not a controlled ablation of quantum advantage [R1].

S2 Earlier binary-grid budget sweep

The historical representation-matched binary custom SNN has 98.73% clean accuracy on the complete 10,000-example official test partition. Its nine-budget attack sweep uses 1,000 clean-correct examples and a different checkpoint, temporal binning and manifest from the controlled four-model study. In particular, its $B_\infty = 1$ ASR is 82.9%, whereas the controlled campaign's custom-model rate is 100.0%. Both values are retained as results of separate experiments [R5].

Table S2. Historical binary-grid attack sweep.

Budget	Successes / N	ASR (%)	Actual B_∞	Actual B_1	Actual B_0
$B_\infty = 1$	829 / 1,000	82.9	1–1	568–2,604	568–2,604
$B_\infty = 2$	1,000 / 1,000	100.0	2–2	840–4,231	573–2,766
$B_\infty = 3$	1,000 / 1,000	100.0	3–3	1,092–5,545	576–2,841
$B_1 = 500$	1,000 / 1,000	100.0	3–9	500–500	268–463
$B_1 = 750$	1,000 / 1,000	100.0	4–9	750–750	364–615
$B_1 = 1{,}000$	1,000 / 1,000	100.0	6–9	1,000–1,000	438–790
$B_0 = 200$	1,000 / 1,000	100.0	8–9	683–955	200–200
$B_0 = 300$	1,000 / 1,000	100.0	8–9	1,051–1,403	300–300
$B_0 = 400$	1,000 / 1,000	100.0	9–9	1,369–1,913	400–400

All realized budget ranges are transcribed from the historical report [R5]; they are not regenerated from attack tensors. Budget families have different units: maximum displacement, total displacement and number of retimed packets.

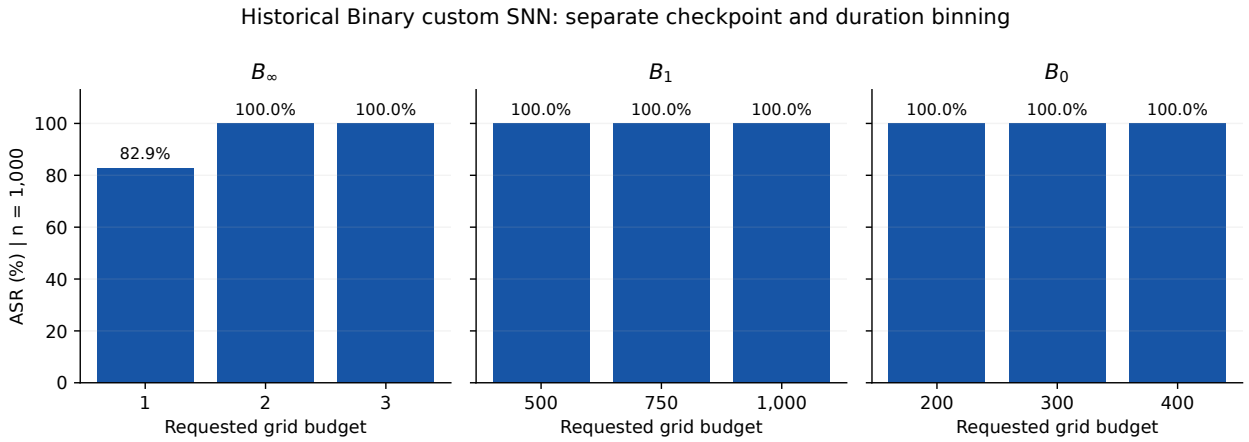


Figure S2. Historical binary-grid success rates for three budget families. Units differ by family. This sweep is not merged with the controlled four-model results [R5].

The corrected integer-grid campaign separately freezes clean-correct manifests using only strict integer-grid predictions. Its status report lists 8 of 33 attack conditions complete and 25 remaining, with the next condition seed 42 and $B_0 = 300$. The preserved final ASR summary is absent. Historical integer attack values are excluded from the corrected campaign summary because they cannot replace its required corrected artifacts [R5].

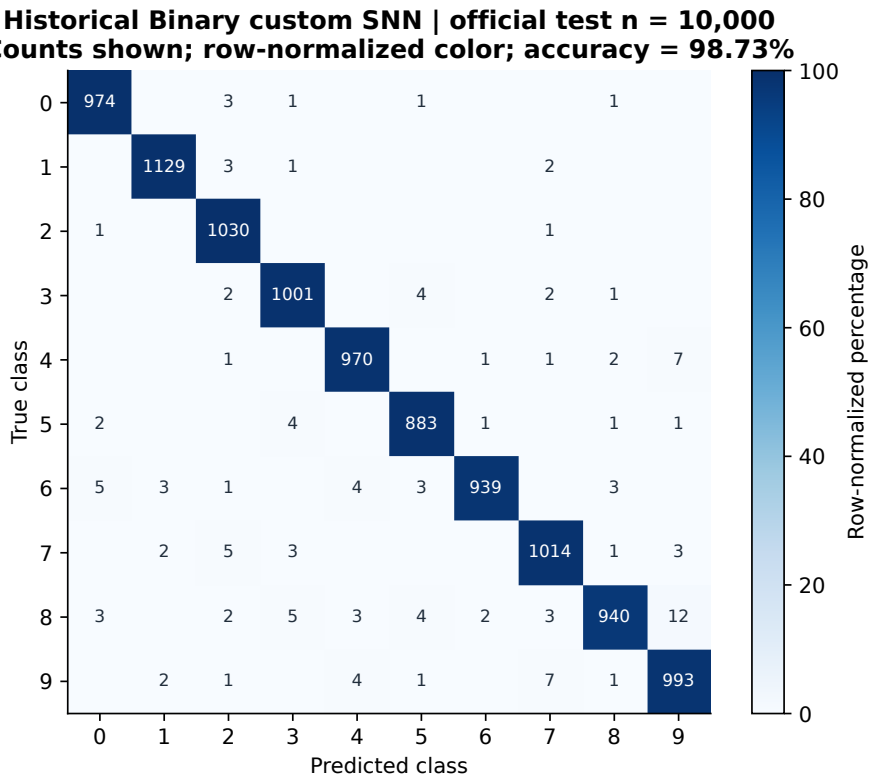


Figure S3. Clean confusion matrix of the historical binary custom SNN on 10,000 official-test examples. The diagonal sums to 9,873, giving 98.73% accuracy. This is not a QSNN-v3 confusion matrix [R5].

S3 Clean neuromorphic baselines

Table S3. Three-seed clean classical baseline accuracy.

Dataset	Representation	Mean ± SD (%)	Evaluation scope
N-MNIST	Binary grid	98.5167 ± 0.2203	Official test
N-MNIST	Integer grid	98.3867 ± 0.2146	Official test
DVS-Gesture	Binary grid	81.3131 ± 2.8930	Official test
DVS-Gesture	Legacy integer	83.0808 ± 4.6289	Normalized counts
CIFAR10-DVS	Binary grid	50.4000 ± 2.0075	Frozen 10% test split
CIFAR10-DVS	Legacy integer	46.2000 ± 6.1733	Normalized counts

Seeds 42, 123 and 777. DVS-Gesture and CIFAR10-DVS rows historically labeled "integer" retain legacy normalized-count preprocessing and are **non-comparable** with strict integer grids [R6].

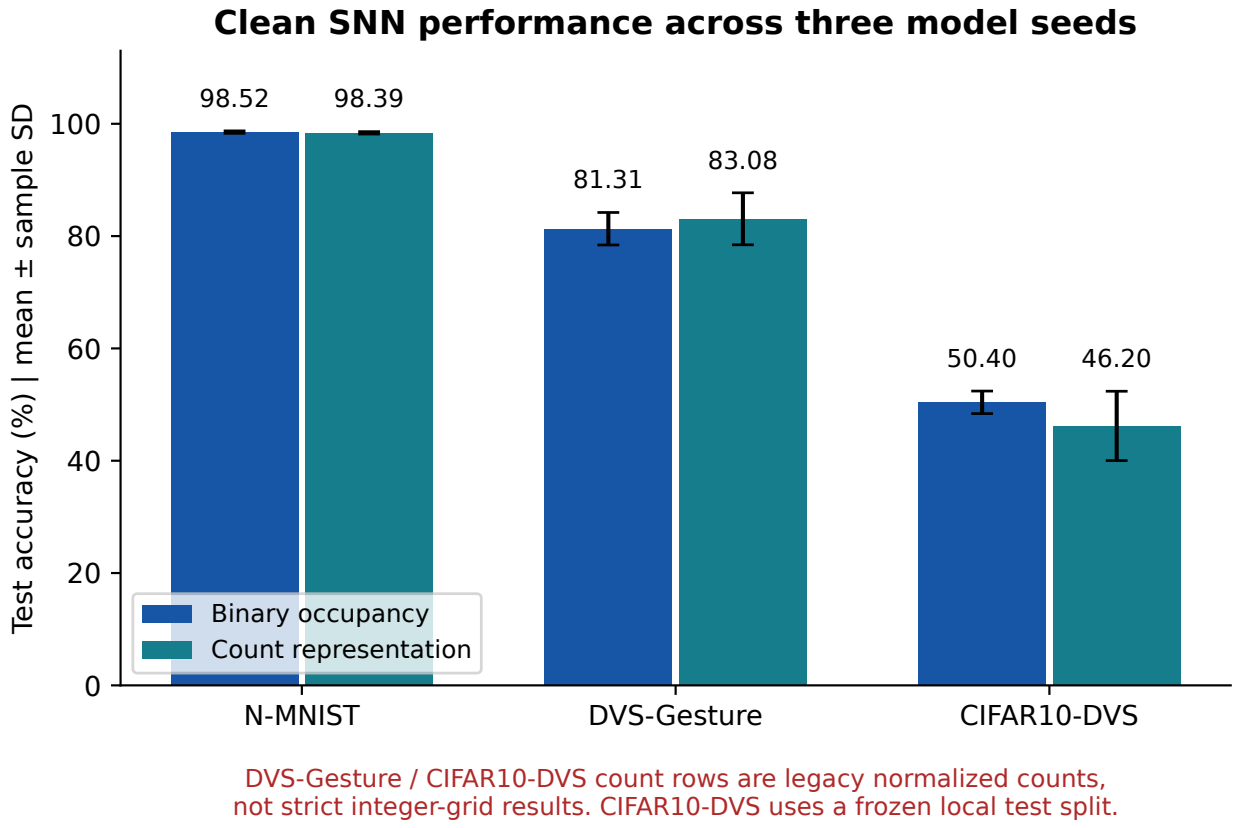


Figure S4. Classical clean accuracy across three seeds. Binary and N-MNIST integer rows are representation-matched. DVS-Gesture and CIFAR10-DVS integer rows retain legacy normalization and do not establish strict-integer performance [R6].

N-MNIST uses all 10,000 official-test examples and DVS-Gesture all 264 official-test examples per seed. CIFAR10-DVS uses a frozen class-stratified 10% split of 1,000 examples because there is no official train/test partition in this protocol. These are clean SNN baselines; they provide no additional QSNN attack or defense results.

S4 Iris attacks and quantum-state diagnostics

Table S4. Historical Iris attack means across five model seeds.

$100\varepsilon/T$	Classical ASR (%)	Adaptive TEMP ASR (%)	Classical $1-F$	Adaptive TEMP $1-F$
2	8.64 ± 6.22	8.64 ± 6.22	0.000987 ± 0.000000	0.000980 ± 0.000006
5	19.18 ± 8.28	19.18 ± 8.28	0.006109 ± 0.000039	0.005766 ± 0.000103
10	32.85 ± 9.67	33.54 ± 10.23	0.024069 ± 0.000276	0.021953 ± 0.000473

Seed-specific clean-correct denominators are 29, 29, 29, 23 and 27. ASR means average seed-specific ratios rather than pooling them. Drift values are reported pure product-state diagnostics [R8].

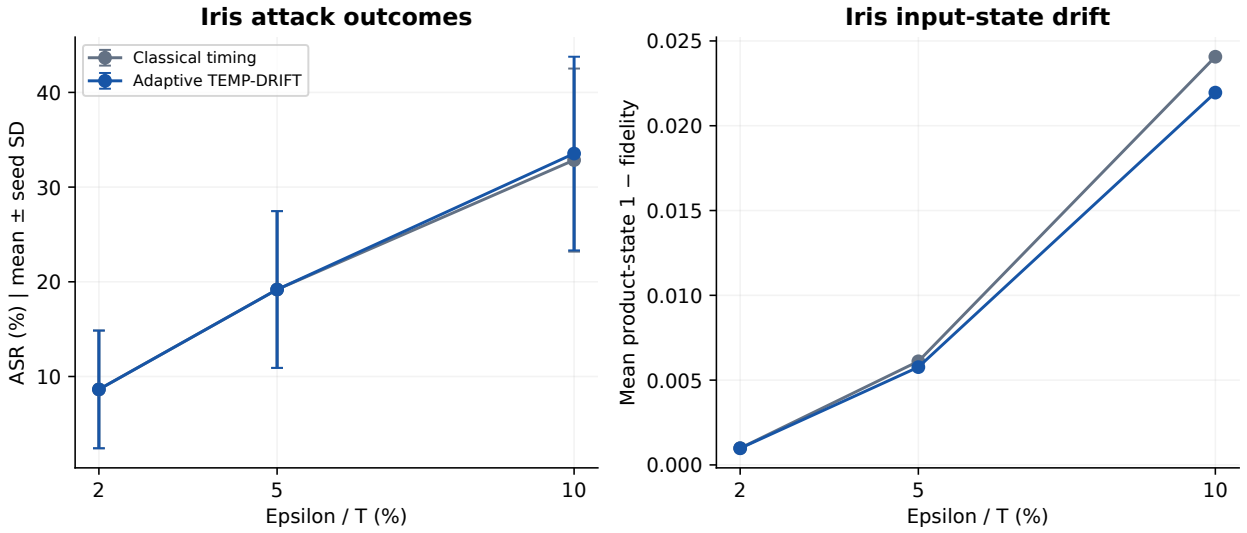


Figure S5. Earlier Iris ASR and state-drift diagnostics. At 10%, TEMP has one extra attack success, but the classical timing attack has larger mean drift. Error bars show between-model-seed variability on small evaluation subsets [R8].

At 10%, classical successes total 46/137 and adaptive TEMP successes 47/137 across the seed-specific populations. These pooled observations repeat the underlying dataset across model seeds, so they are not 137 independent subjects. The exact paired count table has zero classical-only and one TEMP-only success; it does not support attack superiority. The drift quantity refers to the Iris encoding and must not be substituted for an unmeasured N-MNIST circuit-state distance.

S5 Earlier Iris regularization and mechanism gate

Table S5. Earlier Iris regularization replication.

Measure	Baseline	Regularized model	Interpretation
Clean accuracy (%)	84.00 ± 9.25	84.00 ± 8.63	Mean change 0; one seed improves and one degrades
PGD 10% paired ASR (%)	25.58 ± 10.07	24.09 ± 8.45	Two rescued and zero broken across 125 repeated observations
Reference TEMP 10%	—	One rescued observation	Improvement confined to seed 42
Gradient TEMP	—	One rescued observation per τ	No improvement in the other four seeds

Fixed split seed 42; five paired model seeds; $\lambda_q = 5$ and $\lambda_{\text{pred}} = 0$. This is representation regularization, not tomography detection. Source: `results/phase153_results.md`, under [R9].

The Phase-21 confirmatory mechanism gate subsequently fails two of four criteria. Its split-level confidence interval crosses zero, so Stage B is blocked and no candidate defense is trained. An audit PASS in that report means that the negative gate result was handled according to the protocol; it does not mean that the proposed defense succeeded. No hidden-test defense result is inferred from nonexecution [R9].

S6 MNIST development and SHD classical testing

Table S6. Seed-42 MNIST clean validation screen.

PCA dimensions	Explained variance (%)	Validation accuracy (%)	Macro-F1 (%)
16	87.64	86.80	86.75
24	95.42	85.30	85.24
32	98.42	83.55	83.46

Downsampled 8×8 MNIST and a separate classifier; one development seed. Runs have different continuation and timeout histories. No attack or defense conclusion follows [R7].

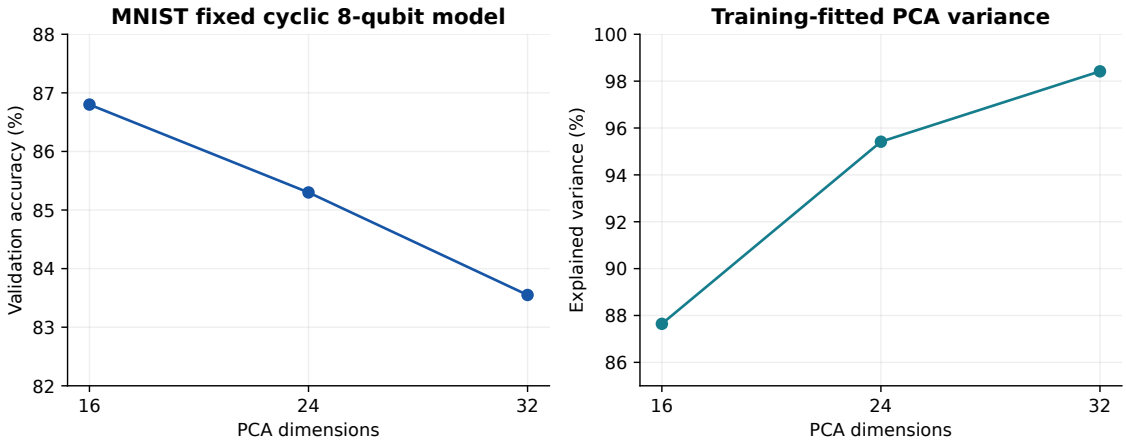


Figure S6. MNIST validation accuracy versus PCA dimension. Retaining more variance did not improve validation accuracy in these development runs. This is not a matched-training-budget ablation [R7].

Table S7. SHD classical SNN official-test re-audit.

Seed	Test accuracy (%)	Macro-F1 (%)
42	65.7686	64.6785
123	69.7880	69.1100
777	66.2102	65.6300
2026	64.0018	63.1983
6543	64.1784	63.9006
Mean $\pm$ SD	65.9894 ± 2.3322	65.3035 ± 2.3123

The historical re-audit reports that recomputed values match its previous test values. This is classical SNN testing on a previously accessed test set, not a QSNN robustness study. Source: `results/shd_snn_test_reaudit_report.md`, under [R10].

S7 Quantum-state notation and monitoring bounds

S7.1 State angles, Bloch vector and density-matrix sign

For an individually encoded one-qubit state, the state-amplitude angle $\theta(t)$ and phase $\phi(t)$ define

$$|\psi(t)\rangle = \cos\theta(t)|0\rangle + e^{i\phi(t)}\sin\theta(t)|1\rangle. \tag{S1}$$

With the usual Pauli matrices, its Bloch vector is

$$\mathbf{r} = \big(\sin 2\theta\cos\phi,\ \sin 2\theta\sin\phi,\ \cos 2\theta\big). \tag{S2}$$

The density matrix and its upper-right entry are therefore

$$\rho = \frac{1}{2}\big(I + \mathbf{r}\cdot\boldsymbol{\sigma}\big), \qquad \rho_{01} = \frac{r_x - i r_y}{2}. \tag{S3}$$

This fixes the conjugation/sign ambiguity in the original appendix.

The repository's Iris angle encoder returns $\theta = \pi t/(2T)$, while a conventional R_Y gate uses rotation angle 2θ. Its amplitude option uses $\theta = \arccos\sqrt{1-t/T}$; its phase option uses $\theta = \pi/4$ and $\phi = 2\pi t/T$. State-amplitude angles and rotation-gate angles must therefore be distinguished. The N-MNIST latent circuit has a separate learned angle map and is not an ensemble average of independent per-event qubits.

S7.2 Squared fidelity and the pure-versus-ensemble distinction

We use squared Uhlmann fidelity,

$$F(\rho,\sigma) = \left[\mathrm{Tr}\sqrt{\sqrt{\rho}\,\sigma\sqrt{\rho}}\right]^2. \tag{S4}$$

For pure product states $|\Psi\rangle = \bigotimes_q |\psi_q\rangle$ and $|\Phi\rangle = \bigotimes_q |\phi_q\rangle$, the implemented squared overlap and trace distance are

$$F = |\langle\Psi|\Phi\rangle|^2 = \prod_q |\langle\psi_q|\phi_q\rangle|^2, \qquad D_{\mathrm{tr}} = \sqrt{1-F}. \tag{S5}$$

A mixed event ensemble instead has the form

$$\rho_{\mathrm{ens}} = \frac{1}{N}\sum_{i=1}^{N} |\psi(t_i)\rangle\langle\psi(t_i)| \tag{S6}$$

and requires the mixed-state fidelity in Eq. (S4). The pure product overlap and mixed ensemble fidelity are not interchangeable computational objects.

A one-qubit state is completely determined by its Bloch vector. Identical Bloch vectors imply identical density matrices and zero quantum Jensen–Shannon divergence; there is no additional phase difference hidden from all three Pauli expectations for a single-qubit density matrix. For eight qubits, identical local marginals do not imply identical joint states. A proposed local monitor must recognize this distinction.

S7.3 A simultaneous finite-shot error bound

For N_{basis} independently estimated observables with outcomes ± 1 and n shots per observable, Hoeffding's inequality and a union bound give

$$\Pr\left[\max_j |\widehat{r}_j - r_j| > \delta\right] \le 2N_{\mathrm{basis}}\exp\left(-\frac{n\delta^2}{2}\right). \tag{S7}$$

Thus a simultaneous absolute error at most δ with failure probability at most η is ensured by

$$n \ge \frac{2}{\delta^2}\ln\frac{2N_{\mathrm{basis}}}{\eta}. \tag{S8}$$

For three one-qubit Pauli bases and a balanced total budget $M = 3n$,

$$\delta = \sqrt{\frac{6}{M}\ln\frac{6}{\eta}}. \tag{S9}$$

At $\eta = 0.05$, this continuous balanced-budget expression gives $\delta \approx 0.536$ for $M = 100$ and $\delta \approx 0.059$ for $M = 8{,}192$. Integer shot allocations use $n = \lfloor M/3 \rfloor$ per basis and slightly increase the bound. These are theoretical conservative simultaneous bounds, not measured detector sensitivity, and do not give full-register tomography complexity.

No final N-MNIST tomography threshold, ROC/AUC, false-positive rate, finite-shot sensitivity, detection latency or adaptive-attack result is available. Monitoring remains a proposal, and the Iris regularization results in Section S5 are a different intervention.

S8 Complete paired timestamp comparison and supporting diagnostics

S8.1 Population, duration convention and computation

All conditions use the same 100 seed-42 validation examples, with ten per class, correctly classified by both frozen models before attack. They are not independent repeat samples across conditions. The timestamp protocol preserves event coordinates, polarity, count, labels and identities while enforcing nondecreasing times and the clean recording endpoints. In timestamp units the implemented duration and displacement constraint are

$$D = t_N - t_1 + 1, \qquad \varepsilon = \varepsilon_f D, \qquad |t'_i - t_i| \le \varepsilon, \quad t_1 \le t'_i \le t_N, \quad t'_i \le t'_{i+1}. \tag{S10}$$

The added unit matches the code's inclusive duration convention; it should not be dropped when reproducing budgets. The ten-bin adapter uses the clean recording endpoints and this duration, with hard accumulation in the forward pass and a surrogate backward interpolation [R2].

PGD scores 21 candidates and repeats 20 inputs for gradients: 41 attack-internal model calls with batch size one and 20 backwards. TEMP evaluates Q proposals in chunks (zero backwards); the number of calls is $\lceil n_0/64 \rceil + 4\lceil n_g/64 \rceil$, not Q. Candidate matching sets $Q = 21$; forward-example matching sets $Q = 41$. Elapsed-time calibration chooses $Q = 4{,}000$ for QSNN-v3 and $Q = 1{,}600$ for SNN, yielding median TEMP/PGD runtime ratios 0.978 and 1.083. Two verification forwards per new record are excluded from attack timing. No backward-to-forward conversion factor is assumed [R3].

Table S8. Complete paired timestamp outcomes. Every row has $N = 100$ joint clean-correct validation examples. Success counts are also ASR percentages.

Condition	Victim	$100\varepsilon_f$	PGD	TEMP	b	c	Both	Neither	Exact p
Original fixed budget	SNN	5	0	0	0	0	0	100	1
Original fixed budget	SNN	10	0	1	0	1	0	99	1
Original fixed budget	QSNN-v3	5	3	4	0	1	3	96	1
Original fixed budget	QSNN-v3	10	3	5	0	2	3	95	0.5
21 candidates	SNN	5	0	0	0	0	0	100	1
21 candidates	SNN	10	0	0	0	0	0	100	1
21 candidates	QSNN-v3	5	3	4	0	1	3	96	1
21 candidates	QSNN-v3	10	3	3	0	0	3	97	1
41 forwards	SNN	5	0	0	0	0	0	100	1
41 forwards	SNN	10	0	0	0	0	0	100	1
41 forwards	QSNN-v3	5	3	4	0	1	3	96	1
41 forwards	QSNN-v3	10	3	3	0	0	3	97	1
Elapsed time	SNN	5	0	0	0	0	0	100	1
Elapsed time	SNN	10	0	1	0	1	0	99	1
Elapsed time	QSNN-v3	5	3	4	0	1	3	96	1
Elapsed time	QSNN-v3	10	3	5	0	2	3	95	0.5

b: PGD-only success; c: TEMP-only success. The original fixed-budget condition is distinct from candidate, forward and elapsed-time matching. Identical aggregate counts do not imply additional independent observations. Exact two-sided paired p-values are checked from the preserved discordant counts [R3].

At 5% on QSNN-v3, the three joint successes and one TEMP-only success give exact paired $p = 1$. At 10% under elapsed-time matching, three joint successes and two TEMP-only successes give $p = 0.5$. No matched condition establishes statistically significant TEMP superiority. Failure to detect a difference is not proof of equivalence.

S8.2 Reported bootstrap intervals

The preserved report gives class-stratified paired bootstrap 95% intervals for the TEMP-minus-PGD ASR difference. The QSNN-v3 5% cells have reported intervals $[0,3]$ percentage points. At 10%, the reported intervals are $[0,0]$ under candidate and forward matching and $[0,5]$ under elapsed-time and original fixed-budget conditions. The SNN 10% original and elapsed-time cells have reported intervals $[0,3]$; all other SNN cells have $[0,0]$. These are report-derived summaries, not intervals regenerated without the original per-example records. A degenerate empirical interval from zero observed discordance is not a population equivalence bound [R3].

S8.3 Attributed realized-distortion diagnostic

The final QSNN status section of `readme_jafar.md` explicitly labels the following summary "QSNN, epsilon=10%, wall-clock matched," immediately after its $Q = 4{,}000$ calibration description. The completed comparison report itself does not reproduce the distortion values, and the recovered per-example arrays needed to verify an exact row linkage are absent. Table S9 is therefore retained as a report-specific supporting diagnostic, rather than a newly validated per-example result.

Table S9. Attributed QSNN-v3 realized-distortion summary for the reported 10% wall-clock condition.

Reported measure	PGD	TEMP	Difference	Relative increase (%)
Normalized timestamp shift	0.0667	0.0925	0.0258	38.7
Fraction of events changing bins	0.6463	0.8907	0.2444	37.8
Frame L_2	105.65	137.85	32.20	30.5

Source: `readme_jafar.md`, final QSNN status, Section 6. Differences and relative increases are descriptive arithmetic from rounded summary values. Aggregation details and per-example uncertainty cannot be reconstructed from the available summary.

The reported bin-change difference is 24.44 percentage points. Larger realized perturbation summaries accompany the report's two extra elapsed-time TEMP successes, but the aggregates do not establish that additional distortion causes those failures. Equal maximum displacement budgets need not yield equal realized distortion. Event-count preservation does not imply invariant time-bin histograms or statistical stealth.

S8.4 Scope of report audit claims

The repository reports 2,400 passing canonical timestamp records and 1,600 completed new budget-comparison records, with reused canonical rows where specified. These are record-accounting statements about the timestamp study, not extra independent attacked subjects and not validation of the separate grid campaign. The budget-comparison README still says that no experiment results exist, while its populated report states completion. Numerical results here follow the completed report [R3]. Restoring underlying arrays, logs, manifests and audit outputs is necessary for full independent reproduction.

The controlled grid report states that all sixteen attack cells passed raw-data reconstruction and prediction audits [R4]; those audits were not rerun here. New static inspection confirms a coefficient of 5 in the pinned upstream L0 loss and a separate capacity penalty of 20 in the local adapter. The report compares the former with 10 in the paper description. Resolved historical execution settings remain unavailable; see Section S11.

S9 Positioning, final-model tables and readout analysis

The following material supports the main timestamp-focused presentation. Relocation does not change the evaluation populations or add experiments.

S9.1 Conceptual literature comparison

Table S10. Conceptual positioning of related work. Entries describe research scope, not a common benchmark or an experimental ranking.

Work	Studied interface or construction	Relation to the present study
Yao et al. [1]	Raw-event attacks with discrete-value optimization and sparsity control	Motivates the raw-event interface; not restricted to our identity-preserving retiming.
Du et al. [2]	Raw-event attacks with configurable latencies	Related temporal attack setting; constraints and budgets require separate specification.
Yu et al. [3]	Capacity-one spike-retiming constraints	Protocol reference for the separate classical packet-grid study.
Jha et al. [4]	SNN features followed by quantum kernels for EEG	Hybrid temporal classification; distinct from our variational probability readout.
Gong et al. [5]	Randomized quantum encodings with specified attacker access	Defense assumptions differ from an attack before our classical frontend.
Dowling et al. [6]	Guarantees for specified quantum-classifier scenarios	Motivates assumption-aware interpretation; not a certificate for our timing budgets.
This work	End-to-end timestamp attacks on QSNN-v3; separate classical grid controls	Paired computational controls, explicit populations and report-grounded evidence.

S9.2 Individual final-model validation runs

Table S11. QSNN-v3 validation on the same 5,000 examples.

Seed	**Accuracy (%)**	**Epoch**
42	97.06	37
123	97.28	24
777	97.52	40
2026	97.44	38
6543	97.06	40
Mean $\pm$ SD	97.2720 ± 0.2119	—

All seeds reuse one pretrained frontend and one split. Reported macro-F1 is $97.2736\% \pm 0.2119\%$ [R1].

S9.3 Complete local packet-grid results

Table S12. Controlled four-model packet-grid results. Clean accuracy and ASR use different stated populations.

Grid	Architecture	Clean test %	ASR B_∞ = 1 %	ASR B_0 = 200 %
Binary	Custom SNN	97.99	100.0	99.5
Binary	ConvNet	96.90	1.9	1.2
Binary	Spiking ResNet18	98.61	0.1	0.0
Binary	VGGSNN	99.21	0.0	0.0
Integer	Custom SNN	98.14	99.9	98.0
Integer	ConvNet	96.58	0.6	1.1
Integer	Spiking ResNet18	98.12	0.2	0.1
Integer	VGGSNN	99.30	0.0	0.0

Clean accuracy uses the complete 10,000-example official test set. ASR uses a separate frozen 1,000-example joint-correct manifest per representation. These are local measurements, not values copied from [3]. Source: [R4].

S9.4 Derivation for the fixed probability readout

For the fixed trained model, let ρ and ρ' denote the clean and retimed quantum states before the fixed readout measurement, and let $q, q' \in \mathbb{R}^{256}$ be the corresponding outcome-probability vectors. Write the logits as $z(q) = Wq + b$, and define the clean pairwise margin against class $k \neq c$ by

$$m_{ck}(q) = (w_c - w_k)^\top q + b_c - b_k. \tag{S11}$$

Here we restrict to the attacked clean-correct subset, so the true class $c = \widehat{c}(q)$ is also the clean predicted class, and w_k is row k of W. Measurement cannot increase trace distance [7, Corollary 3.40], hence

$$\tfrac{1}{2}\|q - q'\|_1 \leq D_{\mathrm{tr}}(\rho, \rho'), \qquad D_{\mathrm{tr}}(\rho, \rho') = \tfrac{1}{2}\|\rho - \rho'\|_1. \tag{S12}$$

The state norm is the trace norm; for probability vectors it is the ordinary ℓ_1 norm. Combining this inequality with Hölder's inequality gives

$$\begin{aligned} m_{ck}(q') &\geq m_{ck}(q) \\ &\quad - 2\|w_c - w_k\|_\infty D_{\mathrm{tr}}(\rho, \rho'). \end{aligned} \tag{S13}$$

Thus a sufficient condition for retaining class c is that the right-hand side remain strictly positive for every $k \neq c$. To see this, subtract the two affine margins and bound the resulting inner product by $\|w_c - w_k\|_\infty \|q' - q\|_1$; positivity of all pairwise margins then preserves the argmax.

This observation separates state displacement from decision failure. A larger state distance need not cause a label change, and a small clean margin can make a smaller displacement consequential. Equation (S13) is a conditional interpretation of the implemented readout, not a certified timestamp radius: no uniform bound from admissible event retiming through the classical frontend to D_{tr} has been established here. The final-model campaign also does not report the per-example states needed to evaluate this condition. Its connection to quantum robustness theory is therefore conceptual rather than an application of an existing certificate [6].

S9.5 Descriptive binomial score intervals

For a reported s successes among n selected examples, let $\widehat{p} = s/n$ and $z = 1.959963984540054$. The plotted 95% Wilson score interval [8] is

$$\frac{\widehat{p} + z^2/(2n) \;\pm\; z\sqrt{\widehat{p}(1-\widehat{p})/n + z^2/(4n^2)}}{1 + z^2/n}. \tag{S14}$$

These intervals are computed from aggregate counts, not recovered per-example outputs. For $n = 100$, the intervals for $s = 0, 1, 3, 4, 5$ are approximately $[0, 3.70]$, $[0.18, 5.45]$, $[1.03, 8.45]$, $[1.57, 9.84]$ and $[2.15, 11.18]$ percent. They summarize a binomial model on the selected subset; deterministic joint-correct selection, repeated conditions and shared examples do not justify coverage claims for the full validation/test population. They neither replace the paired McNemar test nor measure uncertainty of a paired difference.

S10 Static inspection of the preserved attack implementation

This is a source-level audit of the pinned snapshot, not an executed experiment. Code paths are distinguished from proof that particular saved records used them, since most run artifacts are absent.

S10.1 Budget-scaled TEMP schedule

The matched runner explicitly defines the two low-access schedules, rather than truncating the historical 600-candidate initialization. Its candidate evaluator consumes all initialization proposals and all four child batches, with no early stop on success. At each generation it preserves the 12 lowest-margin elites and evaluates only new children. All proposals pass through the timestamp feasibility operator. The final minimum-margin candidate is returned.

Table S13. TEMP schedule defined in the preserved matched runner.

Condition	Initial	Children/generation	Generations	Total Q
Candidate matching	13	2	4	21
Forward matching	13	7	4	41
Elapsed time, QSNN	1,500	625	4	4,000
Elapsed time, SNN	600	250	4	1,600

The source algorithm can be written without implying unmeasured budgets:

1. Read the condition's (n_0, n_g) from the fixed schedule. Draw n_0 uniform perturbations in $[-\varepsilon, \varepsilon]$ per timestamp, apply feasibility, and evaluate their margins.
2. For each $\sigma \in \{0.50, 0.30, 0.18, 0.10\}$, retain 12 lowest-margin candidates. For $j = 0, \ldots, n_g - 1$, use elite j mod 12 as parent, add $0.5(e_a - e_b)$ with independently drawn elite indices a, b, then Gaussian noise of scale $\sigma\varepsilon$ per timestamp. Apply feasibility and evaluate the n_g new children.
3. Replace the pool by the 12 elites plus children; preserve their already computed scores. After the fourth generation return the candidate with minimum margin. Exactly $n_0 + 4n_g$ proposals have been evaluated.

The selected elapsed-time schedules reproduce the stated totals; the generic scaling helper records any rounding adjustment. The report and runner agree on the low-Q definitions, but the saved `budget_definitions.json` and candidate trajectories are unavailable for run-level verification [R3].

S10.2 PGD candidate retention

The preserved `scripts/run_nmnist_common_attack_protocol_seed42.py` scores the clean input and 20 iterates and retains the highest true-label cross-entropy, without a success-priority rule. Cross-entropy ranks true-class probability, while a multiclass decision also depends on competing probabilities; the highest-loss candidate need not be a misclassification even if another iterate is. This establishes a possible retention failure, not its occurrence in the reported records. No success count has been corrected on this basis. A future rerun should record success at every feasible iterate and compare final-selection and any-valid-candidate success under a common documented rule.

S10.3 Timing and inference state

The matched runner requires CUDA and synchronizes immediately before and after `perf_counter` timing. Calibration selects ten joint-clean-correct training examples, one per class, disjoint from evaluation IDs and near each class's median event count among a bounded candidate scan. It uses $\varepsilon_f = 0.10$, two repeats per example, a PGD warm-up and a TEMP warm-up requested at $Q = 400$ (398 actual proposals after rounding). It searches requested budgets

$$Q \in \{800, 1200, 1600, 2000, 2400, 3200, 4000, 4800, 5600, 6400\},$$

recording actual totals after the scaling helper rounds initialization and generation sizes, then choosing the median TEMP/PGD ratio closest to one and accepting a $\pm 10\%$ band. TEMP candidate chunks are at most 64; PGD forwards contain one event example. Evaluation alternates PGD/TEMP order by sample position and excludes two verification forwards from attack

timing. Timestamp tensors use float32 and the quantum backend uses complex64; the attack code does not apply the training loop's float16 autocast. Saved calibration arrays are absent, so absolute times, dispersion and historical device identity cannot be newly verified. The environment note's RTX 4090 observation is a later inspection, not a run-specific timing snapshot.

The timestamp loader calls `eval()` for both victims. The compact SNN and spatial LIF extractor initialize membrane/accumulator tensors inside each forward, with no persistent cross-call neuron state. BatchNorm consequently uses fixed evaluation statistics. The recovered grid reference forwards also explicitly call `functional.reset_net(self)`. Their `PiecewiseLeakyReLU(c=0.0)` surrogate is static evidence about backward choice, not proof of gradient masking. Runtime gradient coverage, alternative surrogates and serial/chunk/order checks remain unexecuted.

S10.4 Unresolved baseline and artifact provenance

The controlled-grid loader, three reference definitions, training script and attack adapters are available at the pinned source commit. Their static findings and remaining runtime linkage are recorded in Section S11; a published architecture citation is not inferred from a class name. No real clean/retimed exemplar or final-model state–margin scatter is constructed without its raw timestamps, probabilities and logits.

S11 Additional provenance and accounting audit

This manuscript revision inspected additional source, not trained victims. The manuscript base is commit `e46a09e89af4`; the historical research source remains pinned to `80d02f5733d4`. Full SHA-256 hashes, exact paths and upstream URLs are in the accompanying `Reports/manuscript_phase4/` manifests. The original evidence manifest and numerical CSVs are preserved.

S11.1 Baseline identity and availability

Table S14. Controlled-grid source identities. Hashes are SHA-256 prefixes of the recovered model files, not checkpoint hashes. Counts marked unverified are not copied from a paper's architecture table.

Report label	Local class	Spiking	Temporal output	Params.	Source hash
Custom SNN	`NMNISTConvSNN`	LIF	Mean of ten per-step classifier outputs	25,482	`be5cc98c0adb`
ConvNet	`SimpleNet_v2`	LIF	Mean of time-major logits via wrapper	Unverified	`88cb8521fcaa`
Spiking ResNet18	`ResNet18`	LIF	Same wrapper	Unverified	`8a40beb21a35`
VGGSNN	`VGGSNN`	LIF	Same wrapper	Unverified	`dd7fc72f3335`

Source-derived custom count: 25,482. Reference runtime counts are unverified in this revision; historical checkpoint-specific counts are unavailable. "Classical" means nonquantum, not necessarily nonspiking.

The loader maps `convnet` to `SimpleNet_v2`, not `SimpleNet_v2_CNN`. All three reference files are byte-identical to `yuyi-sd/Spike-Retiming-Attacks` commit `f16738f80be9`. The wrapper permutes $[B,T,C,H,W]$ to $[T,B,C,H,W]$, requires $[T,B,10]$ outputs and averages logits over T. The reference forwards reset neuron state explicitly. These static observations do not establish order/batch invariance for historical checkpoints or supply an experimental gradient-masking diagnosis.

Table S15. Availability in the supplied local evidence package as of 14 October 2026. Absence here does not assert absence on other branches or external storage.

Artifact	Available	Remaining limit
Source	Models, loaders, attack/training scripts; hashed upstream source	No proof each historical cell executed these exact bytes
Config	Protocol descriptions and draft Phase 3 configs	Resolved historical training/attack config files missing
Weights	No original trained checkpoints	No independent clean/attack inference
Data	Aggregate CSVs and report snapshots	Raw N-MNIST and original caches/manifests absent
Results	Historical aggregate counts and source reports	Per-example trajectories, logits and timing arrays absent
Audit	Attributed historical PASS summaries; new arithmetic/software checks	Original runtime audit outputs not recovered

The budget-comparison README's "no results" statement is stale relative to its populated report. Both originals remain unchanged; a dated reconciliation note accompanies this revision. Numerical values follow the completed report, not the stale status sentence.

S11.2 Test-access risk and fresh-run policy

The recovered controlled-training script chooses the best checkpoint by validation accuracy, breaking ties with loss. Separately, lines 74–79 use final test accuracy to skip or request recipe revision; lines 143–156 set PASS at 0.95 and request revision below it. Thus validation checkpoint selection and possible test-dependent recipe development coexist in the code. No per-run config chronology, training history or test-access log was recovered. The branch establishes a risk, not proof that a historical recipe was changed, and does not justify declaring the historical test untouched.

A new completion-policy utility accepts a completed run independently of test accuracy only when its configuration/source/split identity, validation checkpoint selection, validity/leakage/frozen-recipe gates and checkpoint/history/prediction hashes pass. It is covered by software tests; integration into a full fresh training campaign remains pending. The historical script is preserved. A new campaign must fix recipe and quality criteria on validation, report test outcomes without recipe feedback and retain scientific integrity gates. If the test was used for development, rerunning it does not restore independence; new untouched evaluation data or a defensible external design is needed.

S11.3 Resolved source-level B0 settings

The local batched adapter subclasses upstream `PGDTimeShiftAfterEncoder_L0`. Its explicit constructor values are 40 steps, `alpha_phi=1`, `lambda_cap=20`, temperature 1, no random start, capacity 1 and the requested packet budget. The upstream constructor defaults `lambda_B` to zero, but its forward loss separately hard-codes

$$5\,\mathrm{mean}[\max(S_{\mathrm{soft}} - B_0, 0)] \,/\, \max(1, B_0).$$

Here S_{soft} is the expected number of moved packets. This is the coefficient compared with ten by the historical report; it is not the capacity penalty. The upstream attack-file SHA-256 is

`e2f76fc6b4796d9cece0dbf498ced5f62c4f9581c17476da34ab41a1087e11ea`.

Source resolution is now available; historical runtime objects, overrides and raw attack/audit artifacts are not. The grid remains contextual rather than a causal architecture ranking.

S11.4 Hard-forward adapter and exact paired calculation

For each retimed event define $u_i = 10(t_i' - t_1)/D$. The hard adapter assigns it to $\mathrm{clip}(\lfloor u_i \rfloor, 0, 9)$; the surrogate uses triangular weights $w_{ib} = \max(0, 1 - |u_i - (b + 1/2)|)$ for $b = 0, \ldots, 9$. The straight-through count is $X_{\mathrm{soft}} + \mathrm{stopgrad}(X_{\mathrm{hard}} - X_{\mathrm{soft}})$, followed by the count cap. Hard verification uses the actual classifier argmax and full timestamp validity. Finite differences can be zero strictly inside a hard bin even when the chosen surrogate derivative is nonzero; that observation alone is not an implementation error.

For discordant counts b (PGD only) and c (TEMP only), let $d = b + c$. The exact two-sided conditional convention used here is

$$p_{\text{exact}} = \begin{cases} 1, & d = 0, \\ \min\left(1, 2\sum_{j=0}^{\min(b,c)} \binom{d}{j} 2^{-d}\right), & d > 0. \end{cases} \tag{S15}$$

This states the implementation explicitly; citing McNemar's original framework [9] does not by itself specify an exact-test convention. The marginal Wilson intervals [8] are not paired-difference intervals. Conditions reuse subjects; they are not pooled as independent repeats.

S11.5 Compute units and remaining measurement

"Forward-example count" means the sum of model-call batch sizes; "model calls" counts actual forward invocations. For PGD these are 41 and 41, with 21 scored candidate proposals and 20 backwards. Repeated gradient inputs are computational work. For TEMP the sum is Q_{act} and calls are $\lceil n_0/64 \rceil + 4\lceil n_g/64 \rceil$: 5, 5, 64 and 26 for the four reported schedules. Unique timestamp vectors were not recovered. Two final verification calls per new historical record are separate. No guessed backward conversion or equal-FLOP assertion is made. The requested 400-proposal warm-up actually generates $150 + 4(62) = 398$ proposals. Standalone radius searches are distinct from cumulative best-so-far analyses; prior search and warm-start costs must be included in the latter.

Fresh Phase 3 code records proposals, unique vectors, model calls, sum of batch sizes, backward and verification costs, dtype, device, chunk and separate elapsed times. Its synthetic tests are software checks, not historical timing measurements. Same-device warm-up/synchronized serial-versus-chunk pilots and raw time dispersion remain blocked by missing weights/data. A current hardware inspection cannot replace a historical runtime snapshot.

S11.6 Ablation identity and reproducible entry points

The recovered old QSNN development report covers 154–274-parameter models with local Pauli-Z readout and a different representation. It is not a fixed-frontend classical-head/no-entangler ablation of the final 40,762-parameter, 256-probability model. That controlled experiment, official-test QSNN evaluation, empirical selection audit and real exemplars remain future work.

The package README distinguishes dependency-pinned software checks from model execution. A portable pilot command is `python -m scripts.phase3.runner pilot -config <resolved.yaml>`; it requires real checkpoint and metadata hashes plus a raw-event manifest with provenance. The supplied configuration deliberately leaves these unset and yields blocked status with null metrics. No historical Windows interpreter, absent ResearchLoop contract or unexecuted test unlock is claimed to exist on this system.

S12 Repository evidence map and reporting boundary

All paths below are repository-relative at commit

`80d02f5733d4cbe7c3e69b828dff66d50ba5c211`.

The package includes corresponding report/source snapshots where available; its `data/evidence_manifest.json` records original snapshot URLs and hashes. Reports and derived CSV transcriptions are the same evidence, not independent replications. Most checkpoints, raw per-example arrays, manifests, original logs and audit outputs referenced by the reports are absent. Internal arithmetic verification during manuscript preparation is distinct from a new execution or full experimental reproduction.

[R1] Final hybrid model and validation. Primary report:
`results/nmnist_hybrid_qsnn_seed42/nmnist_qsnn_v3_multiseed/report.md`.
Companion implementation: `models/nmnist_hybrid_qsnn.py` and `scripts/run_nmnist_qsnn_v3_multiseed.py`. Development reports: `results/nmnist_hybrid_qsnn_seed42/frontend_v3_report.md` and `results/nmnist_hybrid_qsnn_seed42/quantum_head_v2_report.md`. Supports architecture, validation-only results, shared pretraining and development history.

[R2] Timestamp attacks and representation.
`scripts/run_nmnist_temp_drift_v2_seed42.py`,
`scripts/run_nmnist_common_attack_protocol_seed42.py`, and
`scripts/run_nmnist_attack_protocol_v3_auditable_seed42.py`.
Supports native-event retiming, margin-based derivative-free TEMP, timestamp projection, adapter and inclusive duration convention. It does not establish a quantum-drift objective for the final attack.

[R3] Corrected computationally matched comparison.
`results/nmnist_budget_matched_comparison_v2_seed42/report.md`; runner:
`scripts/run_nmnist_budget_matched_comparison_v2_seed42.py`.
Supports original fixed-budget and matched paired outcomes, computational accounting, calibration, exact tests and attributed bootstrap intervals. The contradictory stale README is preserved separately at `results/nmnist_budget_matched_comparison_v2_seed42/README.md`. Supporting distortion and audit-accounting summaries come from the final QSNN section of `readme_jafar.md`.

[R4] Controlled four-model packet-grid study.
`Reports/nmnist_controlled_four_models_lr1e4_comparison.md`.
Supports all eight clean rows, sixteen attack cells, common-recipe controls, separate binary/integer manifests and the report-attributed B_0 coefficient discrepancy. Companion reporting audit: `paper_revision_audit/verified_experimental_results.md`. These are local model results rather than published-reference numbers.

[R5] Historical binary verification and corrected integer status.
`Reports/nmnist_seed42_verification.md` and
`Reports/nmnist_integer_corrected_report.md`.
Supports the historical 98.73% binary test result, nine-budget sweep and non-comparability boundary, and the separately incomplete 8/33 corrected integer campaign.

[R6] Clean neuromorphic baselines.
`Reports/clean_accuracy_report.md`.
Supports full-test or frozen-test-split clean SNN results and identifies the legacy normalized-count rows for DVS-Gesture and CIFAR10-DVS.

[R7] MNIST PCA development.
`results/mnist_8x8_pca_dimension_seed42_report.md`.
Supports the single-seed validation screen with unequal continuation histories, not an attack result or matched-budget ablation.

[R8] Historical Iris attacks.
`results/final_attack_report.md`.
Companion notation/metric sources: `encoding/quantum.py` and `metrics/quantum_drift.py`. Supports frozen-checkpoint attack ratios and pure product-state diagnostics, not final N-MNIST circuit-state measurements.

[R9] Iris mechanism gate and earlier regularization.
`results/iris_phase21_scientific_audit.md`,
`results/phase21_results.md`, and `results/phase153_results.md`.
Supports the negative confirmatory gate, nonexecution of the subsequent defense stage and limited earlier regularization observations. Audit PASS denotes protocol handling, not defense success.

[R10] Environment evidence and SHD re-audit.
`paper_revision_audit/hardware_software_versions.md` and
`results/shd_snn_test_reaudit_report.md`.
The environment observations are a September 2026 snapshot, not a saved runtime record for every experiment. SHD values concern a classical SNN and a previously accessed official test set.

The accompanying main paper cites the research literature separately. The bibliography below is shared with that paper; repository evidence identifiers above are preserved for exact artifact provenance.

References

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